

Media Bias ISM

RQ Methods Report

Common Time Windows

- Pre-period: 2015-01-01 to 2019-12-31.
- Post-period: 2020-01-01 to 2025-12-31.

Common Data Inputs

Default database source is configurable with hardcoded fallback:

- Default schema: `public`
- Default table: `articles_no_title_deduped`

Article fields used across RQ pipelines: firm id, publication date, and NewsMTSC scores.

RQ1: Pre/Post Shift in Firm-Level Media Stance

Question. How has media reporting bias toward S&P 500 firms changed after the 2020 recession shock?

Variable Construction

For article a of firm i on date t , with MTSC probabilities (p_a^+, p_a^-, p_a^0) , keep high-confidence observations only:

$$\max\{p_a^+, p_a^-, p_a^0\} \geq \tau, \quad \tau = 0.90 \text{ (default)}.$$

Define article stance component and firm-day outcome:

$$s_a = p_a^+ - p_a^-, \quad \text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} s_a,$$

where $\mathcal{A}_{it}$ is the set of qualifying articles for firm i on day t .

Controls:

- $\text{vol}_{it} = |\mathcal{A}_{it}|$ (firm-day article volume)
- vix_t (daily market uncertainty)

Shock indicator:

$$\text{Post}_t = \mathbb{I}[t \geq 2020-01-01].$$

Baseline Model (Firm FE)

$$\text{bias}_{it} = \alpha_i + \beta \text{Post}_t + \gamma_1 \text{vol}_{it} + \gamma_2 \text{vix}_t + \varepsilon_{it}.$$

β is the main estimand: average within-firm post-2020 shift in stance.

Estimator and Inference

Within transform by firm means, then OLS:

$$\hat{\theta} = (\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top \tilde{y}, \quad \theta = (\beta, \gamma_1, \gamma_2)^\top.$$

Robust HC1 covariance:

$$\widehat{\text{Var}}(\hat{\theta}) = \frac{n}{n-K} (\tilde{X}^\top \tilde{X})^{-1} \left(\sum_{j=1}^n \hat{u}_j^2 \tilde{x}_j \tilde{x}_j^\top \right) (\tilde{X}^\top \tilde{X})^{-1}.$$

Two-sided normal-approx p-values:

$$p_k = 2 \Phi(-|t_k|), \quad t_k = \frac{\hat{\theta}_k}{\widehat{\text{SE}}(\hat{\theta}_k)}.$$

Identification Conditions

1. Conditional within-firm parallel trends after controls.
2. Controls are not post-treatment bad controls in the identification sense.
3. **Stable measurement process:** NewsMTSC score mapping is comparable over time.

Pre-Period Diagnostic and Fallback Trigger

Compute firm pre means:

$$\mu_i^{\text{pre}} = \frac{1}{T_i^{\text{pre}}} \sum_{t \in \text{pre}} \text{bias}_{it}.$$

Track min, max, range, and standard deviation across firms. If range exceeds tolerance (default 0.02), run SCDiD fallback.

SCDiD Fallback (Compact)

- For each firm i , estimate donor weights from pre period:

$$\hat{w}_i = \arg \min_w \|y_{\text{pre}} - X_{\text{pre}} w\|_2^2,$$

then truncate negatives and renormalize.

- Build post synthetic path $\hat{y}_{it}^{\text{syn}} = X_{\text{post},t} \tilde{w}_i$.
- Firm effect:

$$\hat{\tau}_i = \frac{1}{T_i^{\text{post}}} \sum_{t \in \text{post}} (y_{it} - \hat{y}_{it}^{\text{syn}}).$$

- Aggregate fallback effect:

$$\hat{\tau}_{\text{SCDiD}} = \frac{1}{N^*} \sum_{i=1}^{N^*} \hat{\tau}_i, \quad \widehat{\text{SE}} = \frac{s_{\hat{\tau}}}{\sqrt{N^*}}.$$

Fallback Eligibility

A firm is included in fallback only if it has:

- at least 180 observed pre-period days,
- at least 180 observed post-period days,
- sufficient donor coverage for synthetic fitting.